Chapter 1

THE PROBLEM AND ITS BACKGROUND

Introduction

The local coffee shop industry is growing quickly, making the food and beverage market highly competitive. Unlike large cafe chains that rely on expensive and complex cloud software, independent coffee shops usually operate with smaller budgets and simpler daily routines. A major part of their business involves managing items that spoil easily, such as milk, syrups, and coffee beans. Because of this, combining Point-of-Sale (POS) software with automated inventory tracking has become an important area of study in system development. To manage a cafe properly, owners need software that can accurately track raw materials, monitor expiration dates, and link daily sales directly to their backroom storage.

Even though modern software exists, many local coffee shops still track their supplies manually using paper logbooks, clipboards, or scattered Excel files, which are slow to update and prone to human error. Staff must manually count containers, estimate remaining liquid in open bottles, and check expiration dates one by one, adding extra work to daily operations. During busy rush hours, baristas prioritize speed and intuition, which leads to unrecorded spills, over-pouring, and unnoticed spoilage. Over time, the records in these manual systems drift away from the actual stock on shelves. This gap in data leads to

sudden shortages of ingredients, unnoticed waste, and potential food safety risks from expired products.

This progressive gap between recorded and physical stock, referred to in this study as inventory drift, is the central problem BrewManager addresses. Automating deduction alone does not solve it. Even a system that perfectly subtracts recipe quantities at every sale cannot observe what happens outside the transaction: the spilled shot, the over-poured cup, the bottle knocked off the counter, the batch quietly left to spoil. Software can track what was sold, but it cannot see what was lost. A system that only deducts will therefore appear accurate while drifting steadily further from reality, and the longer it runs unchecked, the more confidently it will report figures that are wrong.

Existing research addresses this gap primarily through hardware. Sensor-based and Internet-of-Things inventory systems achieve continuous physical verification, but their cost places them beyond the reach of the independent operators who need accuracy most. This leaves a practical gap between what the literature recommends and what a small cafe can actually deploy, and it is this gap that motivates the present study.

To address the problem of inventory drift without relying on specialized hardware or network infrastructure, this study proposes BrewManager, a mobile Point-of-Sale and reactive inventory management system for local coffee shops. The system operates on a single Android tablet using an embedded SQLite database, an offline-first architecture chosen deliberately because independent shops cannot guarantee stable connectivity and cannot afford for sales to stop

when a connection fails. Within this architecture, BrewManager implements a closed correction loop. Every completed sale automatically deducts the recipe quantity of each ingredient, routing the deduction to the earliest-expiring batch first under a First-Expired, First-Out rule. Because deduction alone cannot account for unrecorded physical loss, a structured end-of-shift audit allows staff to record actual remaining stock through a simple container count and fractional estimate, requiring no scale, no calculation, and no external device. The system computes the resulting variance, writes the correction back to the affected inventory batches, and thereby begins each new shift from a physically counted baseline rather than an accumulated estimate. A deterministic rule-based decision support module completes the loop by identifying ingredients approaching expiration and recommending the menu items that will consume them soonest, converting measured expiry risk into an actionable intervention at the counter.

By making inventory drift visible, measurable, and correctable within the daily workflow of a small cafe, BrewManager offers a low-cost alternative to hardware-dependent inventory accuracy, with the aim of reducing preventable waste, shortening end-of-shift administrative work, and supporting more reliable stock decisions for independent coffee shop owners.

Objectives of the Study

General Objective

To create a mobile Point-of-Sale and reactive inventory management system for independent coffee shops that automates ingredient-level deduction, detects and corrects the variance between recorded and physical stock through a simplified end-of-shift audit, and converts measured expiry risk into actionable recommendations, thereby reducing preventable material waste and manual inventory labor without requiring specialized hardware or network connectivity.

Specific Objectives

1. To develop an offline-first mobile Point-of-Sale module that stores recipe blueprints and automatically deducts correct ingredient quantities upon checkout. The system routes each deduction to the earliest-expiring available batch using First-Expired, First-Out (FEFO) logic with multi-batch splitting, while preserving inventory integrity through a Manager-authorized transaction reversal function.
2. To create a reactive inventory monitoring module that evaluates the shelf-life of every inventory batch against set expiry timeframes, presents the resulting status through color-coded visual alerts, and automatically restricts the availability of menu items whose required ingredients have been depleted.
3. To develop a simplified end-of-shift reconciliation process that measures inventory drift by comparing theoretical stock against a physical count recorded through container tallies and fractional estimation, and that

writes the resulting variance back to the affected inventory batches so that each subsequent shift begins from a verified stock baseline.

4. To implement a deterministic rule-based decision support module that ranks available menu items according to ingredient expiry risk and trailing sales popularity in order to prompt the consumption of near-expiry stock before spoilage occurs, and to record the acceptance of these recommendations for evaluation.

5. To evaluate the system in terms of deduction and reconciliation accuracy, end-of-shift administrative time, and usability, through testing and structured feedback from independent coffee shop staff and owners.

Scope and Limitations

Scope

This study focuses on the design, development, and evaluation of BrewManager, an offline-first mobile Point-of-Sale (POS) and reactive inventory management application. The application is developed exclusively for local, independent coffee shops and is designed specifically as a standalone application for Android tablet devices. To ensure uninterrupted daily operations without reliance on an internet connection, the system utilizes a localized SQLite relational database.

This scope is explicitly limited to a single-branch, single-device deployment and excludes multi-branch corporate dashboards, predictive

analytics, and cloud synchronization. The core functionalities covered in the scope of this study include:

- **Menu and Blueprint Configuration:** A back-office module that allows the manager to perform standard CRUD (Create, Read, Update, Delete) operations for menu items, raw ingredient profiles, and recipe blueprints. A recipe blueprint maps exactly how much of each ingredient is required for a specific beverage, such as the milliliters of milk and espresso in a latte. Records that are referenced by completed transactions are archived instead of permanently deleted, which protects the accuracy of historical sales and deduction data.
- **Inventory Receiving and Batch Registration:** A receiving module in which the manager records incoming stock as inventory batches. Independent coffee shops acquire ingredients through varied channels, including supplier deliveries and retail purchases. As a result, containers of the same ingredient entering the shop at the same time often carry different expiration dates. The system therefore defines an inventory batch by expiration date rather than by receiving event, so a single receiving session can create several batch records. For each combination of expiry date and container size found in a purchase, the manager records the number of containers with that date and the capacity of a single container. Container capacity is stored on the batch rather than on the ingredient, which allows the same ingredient to be held in different

container sizes at the same time. Expiration dates are never grouped or rounded.

- **Sales and Recipe Mapping Engine:** A POS module that processes customer transactions and generates on-screen digital order summaries (e-receipts) instead of printed paper receipts. At checkout, the module automatically deducts the recipe amount of each ingredient using a strict First-Expired, First-Out (FEFO) and multi-batch splitting algorithm. When a transaction requires more than the earliest-expiring batch holds, the engine empties that batch and carries the remainder to the next batch in expiry order until the full amount is satisfied. The module also includes a real-time "declare container lost" override on the cashier screen for mid-shift accidents, such as dropping a bottle. Staff select the affected batch and record how many containers were lost, in quarter-container steps. The system deducts that amount from the batch and logs it as a recorded loss event, separate from the unexplained variance measured during reconciliation. If the deduction empties the batch, FEFO routes the next order to the next available batch. If no stock remains for an ingredient, all dependent menu items are grayed out. To protect data accuracy when an order is recorded by mistake, the module includes a Manager-authorized void function that reverses a completed transaction and returns each deducted amount to the exact batches it came from.
- **Inventory and Expiry Monitoring:** A tracking matrix that checks the shelf-life of every active inventory batch against the device date and

shows color-coded alerts to warn staff of nearing expirations. Each batch is also given a normalized expiry risk score, which is passed to the decision support module.

- **End-of-Shift Reconciliation:** An audit module that measures inventory drift by comparing the system's theoretical stock against a physical count submitted by staff, then applies the correction back to inventory records. Reconciliation is done per batch. Each active batch is audited through two inputs: a count of sealed, unopened containers, and the amount left across all opened containers of that batch, entered in quarter-fraction steps of the batch's registered container capacity. Because the fraction control does not stop at one full container, staff report one combined value no matter how many containers are open. Each control starts at the value entered during the previous shift. The system multiplies the submitted values by the batch container capacity to get the audited quantity, computes the variance against the theoretical quantity, and writes the corrected value directly to the batch. Each shift therefore begins from a verified stock level rather than an estimate that has drifted. Theoretical quantities stay hidden from staff until the audit is submitted. If a submitted count is higher than the total containers recorded for a batch, the system asks for confirmation and flags it as possible unrecorded receiving. If two batches of the same ingredient return variances that cancel each other out, the system flags this as a likely break in expiry-order stock rotation.

- **Decision Support and Waste Prevention:** A deterministic, rule-based algorithm that identifies ingredients close to expiring and determines which available menu items will use them up fastest. These appear as ranked suggestions on the cashier screen, shown together with the expiry condition that triggered them. Menu items are ranked using a fixed weighted scoring matrix that combines the highest expiry risk score among an item's ingredients with its recent sales popularity. Popularity serves as a measure of whether the item is likely to actually sell within the remaining shelf-life. Every suggestion and whether it was acted on is recorded, so the module's effect on spoilage can be evaluated.
- **Descriptive Analytics:** A back-office dashboard that shows material loss over time, drawn from reconciliation logs and filterable by date range and ingredient, so the manager can identify ongoing sources of waste.
- **Manual Data Export and Database Backup:** Because the system runs offline with no cloud synchronization, it provides two ways to preserve data. For review, the manager can export sales, inventory, and reconciliation reports as CSV or PDF files to the tablet's internal storage. For data recovery, a one-tap function copies the complete SQLite database file to external or user-selected storage, and a matching restore function loads a saved database back onto a device. Unlike the report exports, the database copy holds the full state of the system and can be used to rebuild operations after device loss or failure.

- **User Access Control (Permission Matrix):** A single-device PIN-based authentication system that separates user roles. Baristas are given access only to POS operations and blind end-of-shift audit inputs. CRUD configuration, inventory receiving, transaction voiding, and the analytics dashboard are reserved for the Manager account.

Limitations

While the system is designed to improve operational efficiency, the study and the software are subject to the following limitations:

- **Architectural and Hardware Constraints:** The offline-first, single-device design excludes cloud hosting, multi-branch syncing, and remote access. Because backup is started by the user rather than run automatically, data protection depends on the manager backing up regularly. Any activity recorded after the last backup cannot be recovered if the tablet is damaged, lost, or reset. The system does not connect to Bluetooth or IoT devices, including receipt printers. Expiry tracking and FEFO routing also depend entirely on the device's internal clock. The system warns the user when the clock appears to have moved backwards compared to earlier activity, but it cannot check the correct time without a network connection, so a wrongly set clock will reduce routing accuracy.
- **Reconciliation Resolution:** The audit records remaining stock in quarter-container steps rather than exact measured amounts. This is an intentional design choice. It removes any need for weighing equipment, tare weight setup, or density conversion, and applies one consistent

method across liquids, thick syrups, and granular solids. The resulting resolution is one quarter of a container per batch per shift. Because the module is meant to detect drift building up across shifts rather than to measure single servings, this resolution is enough to identify ongoing material loss at the level where purchasing and portioning decisions are actually made. It is not enough to measure the variance of any single transaction, and estimation error is expected to rise slightly when several containers of the same batch are open at once.

- **Dependence on Accurate Receiving Entry:** All expiry monitoring, FEFO routing, and reconciliation calculations come from the batch records the manager creates at receiving. If an expiration date is misread or mistyped, a container capacity is entered wrongly, or a purchase is not recorded at all, that batch will be tracked inaccurately for as long as it exists. The system flags counts that exceed recorded holdings as possible unrecorded receiving, but it cannot detect a batch that was never entered, and it cannot check an entered expiry date against the physical container.
- **Batch Identification in Physical Storage:** Because batches are defined by expiration date, containers of the same ingredient belonging to different batches must be told apart on the shelf. The system depends on staff distinguishing containers belonging to different batches. This is supported at two levels: containers are labeled with their batch identifier at receiving, and stock is arranged in storage so that earlier-expiring batches are

positioned for first access. Where labeling is left out or storage order is not maintained, staff may audit containers under the wrong batch, producing inaccurate batch-level variance even when the total ingredient quantity is correct.

- **Functional and Analytical Boundaries:** The system tracks materials only. It does not handle financial accounting, tax tracking, or employee payroll. It is built for Android tablets and will not run on iOS or web browsers. The system calculates how much material was lost, but it cannot determine why. It cannot tell the difference between spillage, over-pouring, and theft. It detects breaks in expiry-order rotation only when two batches of the same ingredient show variances that cancel each other out. A rotation break that happens alongside material loss of a similar size may not be distinguishable from ordinary waste.
- **Beverage-Centric Architecture and Food Menu Constraints:** The ingredient model, FEFO logic, and reconciliation method are designed for beverage materials such as milk, syrups, and powders held in countable containers of known capacity. The system is not built to fully track prepared food items such as assembled sandwiches or hot meals. These items can still be sold through the POS as whole units, but their preparation, portioning, and waste are not tracked with the same ingredient-level detail as beverages.
- **Beverage Customization Constraints:** The system does not handle per-order modifiers such as ingredient substitutions, add-ons, or

sweetness and strength adjustments. Each menu item deducts one fixed recipe blueprint. Where staff accommodate a verbal customization at the counter without a corresponding menu item, the deduction will not match what was actually poured, and the difference will appear as batch-level variance during reconciliation. Customizations that reduce ingredient usage will register as unexplained surplus, while those that increase it will register as additional loss. Any customization a shop wishes to track accurately must be configured as a separate menu item.

- **Static Decision Support Adaptability:** The decision support module uses a fixed, rule-based weighted scoring matrix. It does not use machine learning or predictive forecasting, so it cannot learn from or adjust to changing seasonal sales patterns on its own.
- **Evaluation Generalizability:** The evaluation of this system is limited to independent coffee shops handling perishable beverage materials. Findings on usability and effectiveness may not apply directly to large franchise operations or other retail types, and testing is limited to the timeframe of the capstone development cycle.

Significance of the Study

The development of the BrewManager system aims to modernize the operational workflows of independent local coffee shop establishments. Specifically, this study will be highly beneficial to the following groups:

Coffee Shop Owners and Management. The system replaces error-prone paper and spreadsheet tracking with automated ingredient-level monitoring. Automated deduction alone cannot account for losses that occur outside a sale, so the system pairs it with a structured end-of-shift audit that measures the gap between recorded and physical stock and corrects the records to match. This gives owners a stock figure that is re-verified every shift rather than one that drifts further from reality over time. Material loss is recorded per ingredient and charted over time, allowing owners to identify which ingredients are consistently lost and to act on patterns that manual tracking leaves invisible. The offline-first local database keeps the POS running during internet outages, while the manual database backup allows records to be preserved and restored if the device fails. The PIN-based permission matrix protects recipes and inventory settings from unauthorized edits and reserves transaction voiding for managerial approval, so sales and inventory records cannot be quietly altered by front-of-house staff.

Baristas and Cafe Staff. By deducting recipe amounts automatically during POS transactions, the system removes the need for manual stock tracking during busy shifts. At closing, the audit asks for only two inputs per batch: a count of sealed containers and the amount left in the opened ones. There is no weighing, no unit conversion, and no arithmetic. Because the fraction control accepts a single combined value for all opened containers, the number of inputs stays the same regardless of how many containers are in use, and each control starts at the previous shift's value so most entries require only a small adjustment. This

directly reduces the cognitive load and administrative stress placed on staff at the end of a shift, allowing baristas to focus on service speed and beverage quality.

Customers. Customers benefit from more consistent product safety and taste. FEFO routing, proactive expiry alerts, and the decision support module work together to ensure that ingredients closest to expiring are used or sold before they spoil, reducing the risk of expired materials reaching a drink. The reactive interface also prevents staff from accepting orders that cannot be fulfilled with available stock, ensuring reliable and transparent service.

Future IT Researchers and Systems Developers. This study provides a reference architecture for closing the gap between recorded and physical inventory in small retail environments without specialized hardware. Existing research addresses this gap largely through sensor-based and Internet-of-Things systems, which remain financially out of reach for independent operators. BrewManager demonstrates that a software-only approach, combining automated deduction with a structured physical audit that writes corrections back to batch records, can maintain inventory accuracy on a single offline device. It further demonstrates that deterministic, rule-based decision support can convert measured expiry risk into a practical intervention without the data requirements of predictive machine learning, offering a model for similar low-resource operational settings.

Definition of Terms

Automated Recipe Mapping Engine. The background process triggered at checkout that reads the recipe blueprint of each ordered item, determines the required quantity of every component ingredient, and issues the corresponding deductions against inventory batches according to expiry order.

Reactive Inventory Management. An inventory approach in which recorded stock levels respond automatically to both recorded sales and verified physical counts, so that the system adjusts its own records in response to measured conditions rather than relying solely on assumed deductions.

Blind Audit. A control applied during end-of-shift reconciliation in which the system withholds its theoretical inventory quantity from staff until their physical count has been submitted and locked, preventing the submitted figure from being adjusted to match the system's expectation.

Container Capacity. The full volume or weight held by a single unopened container of an ingredient, recorded at the batch level so that the same ingredient may be stocked in differing container sizes simultaneously, and used to convert audit inputs into measured quantities.

Inventory Drift. The progressive variance between a system's recorded theoretical inventory and the actual physical stock available, typically resulting from unrecorded events such as micro-spills, over-pouring, or missed manual entries accumulated over time. Also referred to as data drift.

Declare Container Lost. A point-of-sale override enabling staff to record a mid-shift physical loss, such as a dropped bottle, by specifying the affected batch

and the number of containers lost in quarter-container steps. The amount is deducted from the batch and logged as a recorded loss event, separate from the unexplained variance measured during reconciliation.

Decision Support Module. A deterministic, rule-based component employing a fixed weighted scoring matrix to identify ingredients approaching expiration and rank the available menu items that would consume them soonest, presenting these as prioritized suggestions on the cashier screen together with the expiry condition that triggered them.

Descriptive Analytics Dashboard. A back-office reporting interface reserved for the Manager that charts material loss per ingredient over time, derived from reconciliation logs and filterable by date range, enabling review of measurable ingredient waste and identification of recurring loss patterns.

End-of-Shift Reconciliation. The structured audit conducted at the close of each operational shift, in which staff submit a physical count for every active inventory batch, and the system computes the variance against its theoretical quantity and writes the corrected value back to the batch record.

First-Expired, First-Out (FEFO). An inventory management rule requiring that ingredient batches be depleted in order of expiration date, with multi-batch splitting applied automatically when a transaction's requirement exceeds the quantity remaining in the earliest-expiring batch. Where two batches share an expiration date, the batch holding the smaller remaining quantity is depleted first.

Fractional Input. An interface control through which staff record the amount remaining across all opened containers of a batch, expressed in

quarter-container increments. The control is not limited to a single container, so one combined value is submitted regardless of how many containers are open, and each control is initialized at the value recorded in the previous shift.

Inventory Batch. A database record representing a quantity of an ingredient sharing a single expiration date and a single container capacity, holding the number of containers received, the capacity of one container, and the quantity remaining. A single receiving session may produce several batches where expiry dates or container sizes differ.

Offline-First Architecture. A software design in which the application operates entirely on a locally embedded SQLite database within a single Android tablet, making internet connectivity and cloud synchronization non-essential to daily transactional and inventory operations.

Point-of-Sale (POS). The front-of-house module through which baristas process customer orders, record sales transactions, and initiate automated background deduction of ingredient quantities at checkout.

Proactive Expiry Monitoring Matrix. A color-coded interface that evaluates the shelf-life of every active batch against the device date, flags batches according to defined expiry timeframes, and supplies a normalized expiry risk score to the decision support module.

Recipe Blueprint. A configuration record mapping a menu item to the exact quantity of each ingredient required to prepare one unit of that item, forming the basis for automated deduction at checkout.

Suggestion Acceptance. The recorded outcome of a decision support recommendation, indicating whether the suggested menu item was subsequently sold, used to evaluate the module's effectiveness in prompting the consumption of near-expiry stock.

Theoretical Quantity. The quantity of an ingredient the system calculates should remain in a batch, based on its received amount and all recorded deductions, prior to verification against a physical count.

Transaction Void. A Manager-authorized reversal of a completed sale that removes the transaction from sales records and returns each deducted quantity to the specific inventory batches from which it was originally taken.

User Access Control (Permission Matrix). A PIN-based, single-device authentication system enforcing role separation, restricting baristas to point-of-sale operations and blind audit submission while reserving configuration, receiving, transaction voiding, and analytics for managerial accounts.

Volumetric Variance. The measured difference between a batch's theoretical quantity and the physical quantity recorded during end-of-shift reconciliation, representing material loss not accounted for by recorded sales.